Project: Monte Carlo simulation of binary mixtures of Lennard-Jones particles

Objective: To create a Monte Carlo (MC) code for a binary mixture of Lennard-Jones (LJ) particles (types A,B), as argon-krypton or similar.

Outline

1. Write a MC code for a (N,V,T) simulation of a binary mixture of Lennard-Jones (LJ) particles (types A,B). Use $N_A + N_B = 500$ atoms.
2. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.
   - Use the Lorentz-Berthelot rules for cross interactions:

$$\begin{aligned} \sigma_{AB} &= \frac{1}{2}(\sigma_A + \sigma_B) \\ \varepsilon_{AB} &= \sqrt{\varepsilon_A\, \varepsilon_B} \end{aligned}$$

   - Use σ_{AB} and ε_{AB} for the reduced units.
3. Compute the interaction energy U^* and the pressure P^* of the system.
4. Calculate the radial distribution functions g(r) of A-A, B-B and A-B pairs.
5. Compute the running coordination numbers via integration of the g(r)'s.

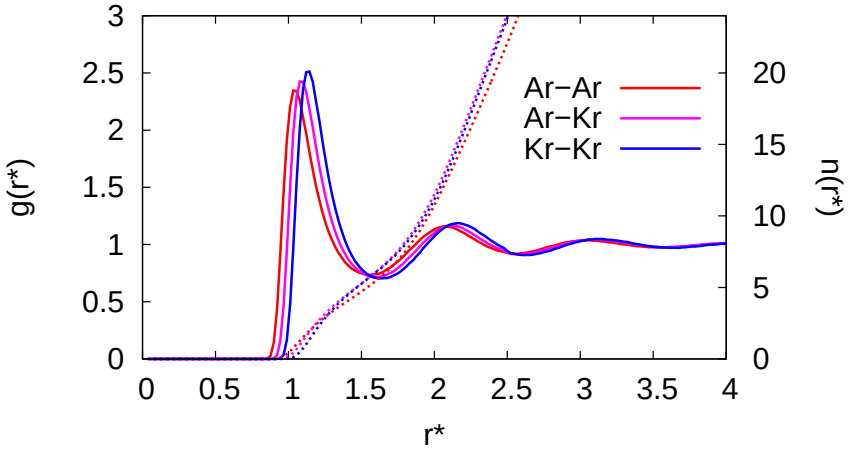
State points of the mixtures (x_{Ar}: is the Ar partial molar fraction)

mixture	x_{Ar}	T*	ρ^*
Ar-Kr	0.485	1.00	0.712
Ar-Ne	0.499	3.973	0.306
Ar-Xe	0.5016	1.700	0.583

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\mathring{A}$)
Ne	20.180	47.0	2.720
Ar	39.948	119.8	3.405
Kr	83.798	167.0	3.67
Xe	131.293	257.4	3.924

Some results to compare with:



Questions

1. **Summarize the Metropolis algorithm for (N,V,T) simulations. Describe the main hypothesis involved in its derivation.**

2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

3. Describe how did you fix the Δ parameter for the trial moves.

4. Derive the expressions for the tail corrections to the energy and pressure ($U_{LJ}^{*,tail}$, $P_{LJ}^{*,tail}$) for a Lennard-Jones binary mixture.